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NUMERICAL DIFFERENTIATION IN EDGE DETECTION FOR IMAGE PROCESSING

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INTRODUCTION

- Image processing involves analyzing and manipulating digital images using mathematical techniques.
- Edge detection is used to identify object boundaries by detecting changes in intensity.
- It is widely used in applications like medical imaging, computer vision, and object recognition.

Original Image



Edge Detected Image





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PROBLEM STATEMENT

- Edge detection in digital images is difficult due to noise and distortions.
- Variations in lighting and contrast affect the accuracy of edge identification.
- There is a need for efficient numerical differentiation techniques to improve edge detection accuracy.



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MATHEMATICAL BACKGROUND

- Images are represented as 2D functions $f(x, y)$, where intensity varies with position.
- Edge detection is based on calculating the first derivative (gradient) of the image.
- Numerical differentiation uses finite difference methods like forward, backward, and central differences.



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MODULE 1:EDGE DETECTION USING PYTHON

- Python is widely used in image processing due to powerful libraries like OpenCV, NumPy, and Matplotlib.
- Images are read and converted into numerical arrays for processing.

```
atm.py | README.md | autorun.inf | app.py.t
File Edit View
# Show images

plt.figure(figsize=(10,6))

plt.subplot(2,2,1)
plt.imshow(image, cmap='gray')
plt.title('Original Image')
plt.axis('off')

plt.subplot(2,2,2)
plt.imshow(sobel_x, cmap='gray')
plt.title('Sobel X')
plt.axis('off')

plt.subplot(2,2,3)
plt.imshow(sobel_y, cmap='gray')
plt.title('Sobel Y')
plt.axis('off')

plt.subplot(2,2,4)
plt.imshow(sobel_combined, cmap='gray')
plt.title('Combined Edge')
plt.axis('off')

plt.tight_layout()
plt.show()
```



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MODULE 2: Edge Detection System for Image Processing

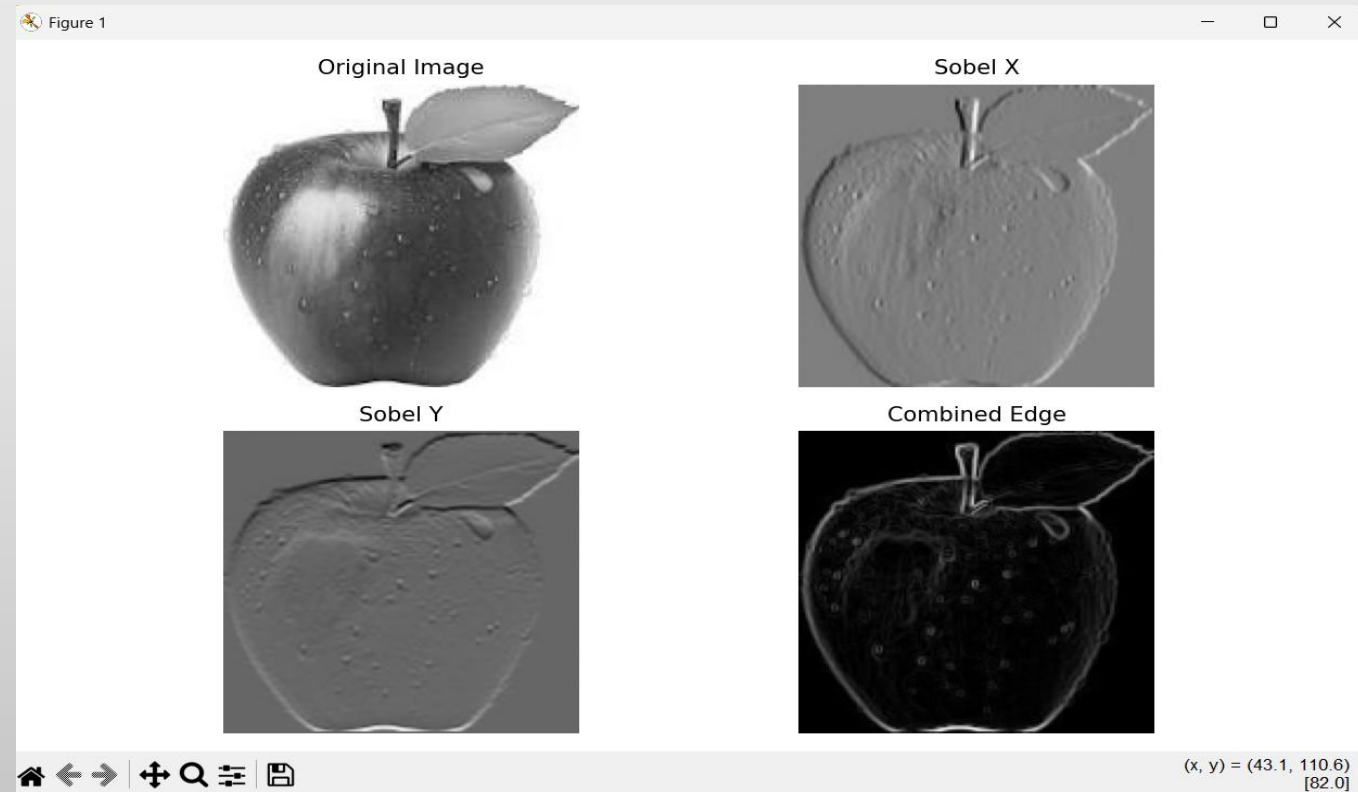
```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

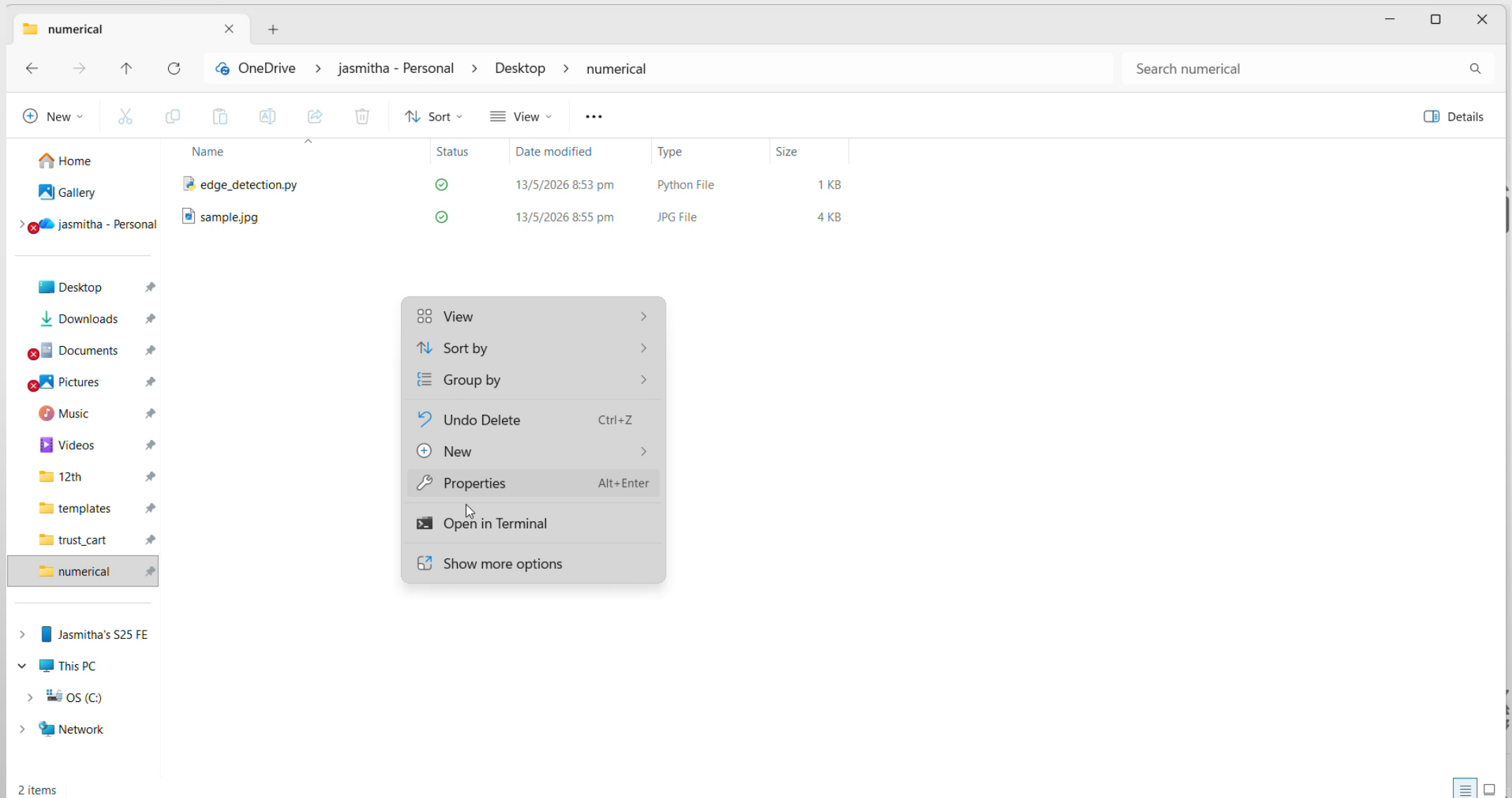
PS C:\Users\jasmi\OneDrive\Desktop\numerical> python edge_detection.py
```

MODULE 3: Performance Analysis of Edge Detection Techniques

- The accuracy of edge detection is evaluated using sample images.
- Results are summarized to identify the most effective technique.



IMPLEMENTATION





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ADVANTAGES

- Improves edge accuracy by detecting sharp changes in pixel intensity more precisely.
- Simple and easy to implement using programming tools like Python.
- Flexible and widely applicable in fields like medical imaging, computer vision, and pattern recognition.



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DISADVANTAGES

- Less accurate for smooth regions – gradual changes in intensity may not be detected properly.
- Amplifies errors – numerical differentiation can increase small errors in pixel values.
- Requires preprocessing – techniques like filtering are needed to reduce noise before applying it.



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CONCLUSION

- Numerical differentiation is an effective method for detecting edges in digital images.
- It helps in identifying sharp intensity changes, which represent object boundaries.
- However, it is sensitive to noise, so preprocessing techniques are important.
- Overall, it plays a key role in image processing applications like medical imaging and computer vision.



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THANK YOU